SHREE GAYATHRI GNANASEKAR

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PROFESSIONAL SUMMARY

Software Engineer with a Master's in Computer Science and experience building full-stack applications, REST APIs, and data pipelines. Proficient in Python, Java, JavaScript/TypeScript, and C++, with a strong background in writing maintainable code and implementing automated testing. An analytical problem-solver adept at collaborating in cross-functional teams to deliver scalable, reliable software.

EDUCATION

Master of Science, Computer Science | *Texas State University* — **Aug 2024 - May 2026 (Graduated)**
Bachelor of Engineering, Computer Science and Engineering | *Avinashilingam Institute* — **Sep 2020 - May 2024**

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++, C#, Kotlin, R
Frontend: React, HTML5, CSS3, WebSockets
Backend: Node.js, Express.js, FastAPI, Flask, Django, REST APIs, Microservice, Microsoft .NET
Databases: SQL, MySQL, PostgreSQL, Firebase, Oracle (PL/SQL), NoSQL
Cloud & DevOps: AWS (S3, EC2), Docker, GitHub Actions CI/CD, Azure, AWS Cloud Practitioner Essentials Certification
AI & ML: LLM Applications, AI Agents, RAG Systems, Model Evaluation, Prompt Engineering, PyTorch, TensorFlow, Keras, Pandas, NumPy, SciPy, OpenAI API, Amazon Bedrock, Google AI Professional Certification
Systems & Performance: Multithreading, GPU Acceleration (CUDA/OpenCL), Parallel Computing, Computer Vision
Tools: Git, Unity, VS Code,Cursor, IntelliJ, Magic Leap Hub, Jupyter Notebooks

PROFESSIONAL EXPERIENCE

Texas State University | *Research Assistant* — **Aug 2025 - Apr 2026**
- Built Mixed Reality (MR) modules for electrical hazard awareness in Unity (C#) for Magic Leap 2 using spatial mapping and depth sensing
- Integrated REST APIs and AI/LLM services into real-time ETL pipelines using Python, Pandas, and SciPy, reducing inference latency by 20%, validated through end-to-end data analysis in Jupyter Notebooks with metrics visualized in Power BI
- Improved MR modules stability by 30% through profiling, structured logging, modular refactoring, and continuous integration testing
- Prototyped LLM-powered agent workflows and evaluated system performance through iterative experimentation and data-driven analysis
- Analyzed experimental data using SciPy and Jupyter Notebooks to derive actionable insights for project improvements

Gateway Software Solutions | *Web Developer Intern* — **Jul 2023 - Dec 2023**
- Developed a full-stack microservice application using React, Node.js, and MySQL, integrated build and deployment workflows, and deployed through CI/CD pipelines
- Built and maintained RESTful APIs and third-party integrations supporting scalable business workflows
- Collaborated in Agile sprint cycles to deliver new features, reducing production defects by 30% and improving page-load performance by 20%

National Small Industries Corporation | *Android Application Developer Intern* — **Jul 2022 - Dec 2022**
- Developed Android apps in Java/Kotlin with Firebase (auth, real-time DB, analytics) for secure data management
- Improved app stability by 25% through structured logging, error handling, and debugging workflows
- Optimized API response times and feature adoption via targeted performance improvements

PROJECTS

ProArcade - AI-Powered Gamified Productivity Platform | <u>https://github.com/shree2402/ProArcade</u> — **Jun 2026 - Present**
- Built a cross-browser full-stack productivity platform compatible with Chrome, Edge, Safari, and Firefox using React, TypeScript, Node.js, PostgreSQL, and Prisma with JWT authentication and persistent game sessions.
- Integrated AWS S3 and Amazon Bedrock (Claude 4.6 Sonnet) to perform multimodal AI-based image verification and secure media storage.
- Designed scalable REST APIs, task management workflows, and cloud-based architectures supporting personalized task assignment and game progression.
- Implemented monitoring and logging for backend services using AWS CloudWatch to support integration and maintenance of system processes.
- Documented REST API specifications and prepared technical support guidelines to streamline troubleshooting and future enhancements.

AR-Enhanced E-Commerce Web Application with Virtual Try-On | *Texas State University* — **Feb 2026 - May 2026**
- Developed an AR-enabled full-stack e-commerce application using HTML5, CSS3, JavaScript (ES6), Bootstrap 5, and Supabase, supporting 100+ products
- Integrated Hugging Face API, WebXR, and Google Model Viewer to deliver AI-powered virtual try-on and real-time product visualization
- Designed scalable database workflows managing 1000+ records and leveraged Cursor IDE/Codex for building, testing, and debugging

Eye Tracker System | *Texas State University* — **Sep 2025 - Oct 2025**
- Engineered a real-time eye-tracking system in C++ using Computer Vision and Gaze Estimation, achieving 25ms latency for pupil detection and tracking
- Optimized performance via CUDA/OpenCL GPU acceleration, multithreading, and efficient memory management to handle real-time video streams

QAAsker – Consistency Testing Framework | *Texas State University* — **Aug 2025 - Dec 2025**
- Evaluated 3 NLP question-answering models across 1,000+ samples using metamorphic testing to analyze robustness, consistency, and model behavior
- Built an end-to-end testing pipeline with PyTest and GitHub Actions CI/CD
- Developed a weighted voting ensemble to improve answer reliability and robustness analysis for large language models

PUBLICATIONS

A Deep Learning Approach for Real-Time Sign Language Detection and Response Through Avatar Model - *IJRASET, 2024*

CERTIFICATIONS

- AWS Cloud Practitioner Essentials
- Google AI Professional Certificate